

1) Find out the history of Tower of Hanoi

A Tower of Hanoi was invented in 1883 by French mathematician Edouard Lucas.

→ The puzzle consists of

- (1) Three rods.
- (2) A no. of disks of diff sizes stacked on one rod.
- (3) The goal is to move the entire stack to another rod following rules:
 - (i) Move only one disk at a time
 - (ii) Only the top disk of a stack may be moved.
 - (iii) Larger disk is not allowed to place on smaller disk.

→ It became an important example mathematics and Computer Science because it demonstrates

- (1) Recursive problem solving
- (2) Mathematical Induction
- (3) Algorithm design

→ Lucas attached a fictional story i.e. Tower of Brahma

→ In temple, 64 disks are moved by monks b/w 3 diamond pegs.

→ When they complete the task, the world will end.

DSA observations in Surroundings

1) Alarms on Mobile
→ Multiple alarms ring one after another.

2) Phone Contacts
→ Contacts are stored in an organised manner making searching faster

3) College Bus.

→ Students boards one after another, those who get first get the seat first

4) Classroom Attendance:-

The attendance registry stores students in a sequential list.

5) College Canteens:-

Students wait in line, student in queue is served first.

6) Web Browser:-

Back button returns to the previously visited page.

7) Google Maps:-

Maps calculate shortest and fastest route to destination.

8) Whatsapp Messages:-

Messages are display in the order they are sent.